COVID-19 Cohort Ward Pathway

Pathways to cohort wards

All admissions to C-19 need to be discussed with COVID-19 medical consultant.

ED have direct admitting rights to Amber Aran but need to communicate decision to admit there to ensure prompt review (currently Level 1 patients only).

1. WAST and GP admissions via ED triage.
2. Presentation from another area within the Hospital – prior discussion with COVID-19 medical consultant on call.

ONLY ADMIT THOSE WITH RESPIRATORY COMPROMISE

Question number 1:

Does this patient need admission?

Home to self-isolate is the default position.

Question number 2:

Does the patient meet the case definition for an inpatient case (PHW/PHE)?

Clinical/radiological evidence of pneumonia or ARDS

(YG team guidance note: are they hypoxic with sats<94% on air and is this new?)

OR

Flu-like illness with Temp ≥ 37.8^{O}C plus one of the following which must be acute (YG team guidance note: >14 days of symptoms in this context does not class as acute): persistent cough, hoarseness, nasal discharge, nasal congestion, shortness of breath, sore throat, wheeze, sneezing.

(NB atypical presentations in the immune-suppressed or elderly).

COVID-19 cohort ward assessment

ED will have done an initial assessment comprising:

Brief clinical and social history (use ED proforma and blue clerking proforma but keep paper in clean area on main ward).

Bloods (FBC, U&E, LFTs, CRP, CK, Coag, LDH, Ferritin, Troponin, Quantitative D-Dimer.....or once available COVID-19 WCP request set).

ABG if sats < 94% on room air.

Throat swab X 1 if planning to admit.

Sputum for COVID-19 PCR if actively producing sputum (↑ sensitivity but do not induce).

ECG

CXR

The options post-assessment:

1. Home with **safety netting advice**.
2. C-19 ward/Amber Aran ward.
3. General medical ward (does not meet case criteria but need admission for other reasons).
4. ITU.

Community onset			Hospital onset
Group 1	**Group 2**	**Group 3**	**Group 4**
Asymptomatic or mild symptoms without dyspnoea, <70, without adverse prognostic factors and negative CXR.	Mild or moderate symptoms including dyspnoea, CXR with pneumonia or mild symptoms with adverse prognostic factors	Severe pneumonia with respiratory failure/ARDS or hemodynamic instability	Variable severity of disease; may be complicated by HAP
Consider home to complete self isolation	May be able to discharge following assessment	Hospital admission and consideration of intensive care if for escalation	

For anything other than home additional clinical and social history to be clarified and clinical frailty score documented as part of a holistic frailty assessment (see NICE guidelines).

If the decision is admit please refer to COVID-19 cohort ward pathway below.

Early decisions, discussions and documentation around escalation status are needed.

COVID-19 cohort ward pathway

Airway

If compromised and for escalation immediate transfer to the negative pressure facility if available OR if not a side room, call 2222 and commence airway manoeuvres (NB this is an aerosol generating situation and PPE needs to be reflect this). See appendix for RCP statement.

Otherwise if possible keep the fluid-repellent surgical mask on the patient.

Breathing

<table>
<tr><th></th><th>Type 1 Respiratory Failure
PaCO2 < 6kPa</th><th colspan="2">Type 2 Respiratory Failure
PaCO2 ≥ 6kPa</th></tr>
<tr><td>Target Sats
(prescribe on drug chart)</td><td>94 – 98%</td><td colspan="2">88 – 92%
(also if history of COPD/otherwise at risk of hypercapnia e.g. chest wall deformity/known neuromuscular diagnosis)</td></tr>
<tr><td>Titration</td><td>Escalate O2 as follows:
1. Nasal cannulae 2-4L/min
2. 35% Venturi
3. 40% Venturi
4. 60% Venturi
5. NRB 10-15L/min</td><td>pH ≥ 7.35

Escalate O2 as follows:
1. N/C 2-4L/min
2. 35% Venturi
3. 40% Venturi
4. 60% Venturi
5. NRB 10-15L/min</td><td>pH < 7.35

Contact COVID-19 medical/critical care on call for consideration of ventilatory options</td></tr>
<tr><td>Monitoring</td><td>Repeat ABG if Fi02 needs escalating (especially if needing FiO2 ≥ 60%) and calculate PaO2/FiO2 ratio (see box below).</td><td colspan="2">Repeat ABG after one hour in target sats range and if FiO2 needs escalating.</td></tr>
<tr><td>Escalation</td><td colspan="3">If for escalation to critical care (level 3) for intubation contact COVID-19 critical care on call for early intubation.

<u>Refer to ICU if PaO2/FiO2 < 27 kPa at any FiO2 or a pO2 of less than 15 kPa on 60% oxygen.</u>

If there is any doubt about escalation plan please discuss early with COVID-19 medical/critical care consultant on call.

If for immediate intubation this should be done in the negative pressure facility (or on a ward side room if no negative pressure facility available). Ideally intubations should be done in ITU setting.

Intubation and Cardiopulmonary resuscitation are both aerosol generating procedures and full PPE is required.

NIV/CPAP are also considered aerosol generating and should only be commenced following discussion with COVID-19 medical/critical care on call. Full PPE is required during their use and for one hour after cessation.

For patients on domiciliary NIV/CPAP please discuss with COVID-19 medical on call.</td></tr>
</table>

If the patient is for escalation to critical care consideration of early intubation is paramount before starting respiratory support other than face-mask oxygen on cohort wards.

1. Needs intubation – straight to ITU.
2. Suitable for level 3 but not needing immediate intubation – CPAP/NIV in the ITU environment, including their escalation areas. If over-capacity CPAP/NIV delivered by medical teams.
3. Not suitable for level 3, with a defined escalation of level 2 only – CPAP/NIV delivered by medical teams.
4. Not suitable for level 2 care but for active treatment – ward based care delivered by medical teams.

If CPAP/NIV required refer to local guidelines.

There is no role for nasal high flow.

Circulation

Be conservative with IV fluid replacement if patient not clinically fluid deplete due to risk of developing ARDS.

Consider whether 'sepsis six' guidelines apply. Remember the golden hour in this situation for antibiotic administration.

Disability

Document AVPU/GCS on blue clerking proforma and is airway therefore at risk?

Escalation plan

ALL PATIENTS ADMITTED SHOULD HAVE AN ESCALATION PLAN

Cardio-pulmonary resuscitation is an aerosol generating situation and PPE considerations apply (See appendix for RCP statement). It is of paramount importance that resuscitation status is decided, discussed and documented. There are twice daily handovers between medical and critical care COVID-19 on call consultants.

The resuscitation team needs FIT tested PPE, visor, a full fluid repellent surgical gown and extended cuff gloves for all. The safety of each individual needs to be safeguarded before they assist in the resuscitation attempt.

Don't ever forget Glucose

Four hourly BMs once admitted to the cohort ward(risk of DKA).

Other considerations

Prophylactic LMWH as standard.

Consider GI prophylaxis with H2-antagonist/proton pump inhibitor.

AVOID the use of NSAIDS. **AVOID** corticosteroids unless needed for a co-morbid condition.
AVOID the use of fans for cooling.

Empirical antibiotics are indicated pending PCR result and approximate to local guidelines. Choice should be guided by CURB 65 score and severity grouping (see below). If using a macrolide check QTc interval and avoid if QTc>450 in a male or QTc>470 in a female.

Community Onset			**Hospital Onset**
Group 1 *Equates to non-pneumonic LRTI on Betsi Guidelines*	**Group 2** *Equates to CURB 65 0-2 CAP on Betsi Guidelines*	**Group 3** *Equates to CURB 65 3+ CAP on Betsi Guidelines*	**Group 4** *Variable severity*
Clarithromycin 500mg PO BD 5 days **OR** Doxycycline 200mg stat then 100mg OD 5 days	Amoxicillin 500mg PO TDS (and Clarithromycin 500mg PO BD if CURB 65 2) 5 days **OR** Doxycycline 200mg stat then 100mg OD 5 days	Tazocin 4.5g IV TDS and Azithromycin 500mg PO OD (IV route if PO unavailable) 5 days **OR** <u>Penicillin allergic:</u> Co-trimoxazole 960mg PO BD (IV route if PO unavailable)and Azithromycin 500mg PO OD (IV route if PO unavailable) 5 days	<u>Non-severe</u> Doxycycline 200mg stat then 100mg OD 5 days **OR** <u>Severe</u> Tazocin 4.5g IV TDS and Azithromycin 500mg PO OD (IV route if PO unavailable) 5 days **OR** <u>Penicillin allergic:</u> Co-trimoxazole 960mg PO BD (IV route if PO unavailable)and Azithromycin 500mg PO OD (IV route if PO unavailable) 5 days

Other microbiological samples: Urine MC&S/pneumococcal and legionella urinary Ag. Blood cultures. Stool cultures.

Daily bloods on COVID-19 cohort ward (see flow chart and use the WCP COVID-19 blood work set once available).

CXR if oxygen requirements increasing.

Manage co-morbid conditions/other physiological considerations

And be in liaison with colleagues by phone: e.g. the dialysis patient, e.g. the pregnant patient, etc.

Cohorting considerations

Try to distance the first admissions in different bays. IP&C will provide guidance.

Once there are confirmed cases cohort them away from unconfirmed cases.

As the number of confirmed cases grow they will take over the cohort wards in blocks.

For those confirmed or unconfirmed cases that are aerosol generating, try to cohort in a side ward or single bay.

The cubicles should otherwise be reserved for those who are at extreme risk or who pose an extreme risk to others: e.g. unconfirmed neutropenic patients, those with a convincing story for another communicable disease such as gastroenteritis and IF POSSIBLE in a peri-arrest situation where CPR is anticipated to be required.

Infection prevention & control

For decontamination procedures refer to Domestic Services/IP&C SOPs.

Discharge from COVID-19 cohort wards

When a patient confirmed/unconfirmed is well enough for discharge home this is the preference if social circumstances allow.

ECG pre discharge and compare with admission ECG (risk of viral myocarditis and therefore cardiomyopathy).

Liaise with the IP&C team about when infection control precautions can be stepped down and discharge to ward can be considered (open bay/side ward).

References

Novel coronavirus (COVID-19): Advice for the NHS in Wales: Public Health Wales statement 13.3.20

COVID 19 Cohort plan for Ysbyty Gwynedd (Acute Medicine/Respiratory) Patients presenting at Emergency Department: PHASE ONE: circulated 13.3.20

Clinical management of severe acute respiratory infection when COVID-19 disease is suspected: WHO 13.3.20

BTS Emergency Oxygen guidelines: 2017

All Wales COVID 19 guidelines

NICE rapid COVID 19 critical care adult guidelines

RCP statement

CPR; personal protective equipment and COVID-19

Details

Date: 20 March 2020

We are aware that healthcare workers are facing difficulties accessing the right PPE for caring for people with suspected or confirmed COVID-19.

It is important that we all follow the PHE guidance for infection prevention and control and Resuscitation Council UK guidance on CPR and resuscitation.

In particular, **do not begin CPR on a patient with suspected or known COVID-19 until you are wearing at least the locally/nationally agreed minimum level of PPE.**